INDOOR VERTICAL GARDEN MONITORING AND AUTOMATION SYSTEM

ABSTRACT

Indoor vertical gardening offers a practical approach to crop cultivation in limited spaces; however, maintaining suitable environmental and nutrient conditions requires continuous monitoring and appropriate control. This project proposes an Indoor Vertical Garden Monitoring and Automation System for hydroponic lettuce (Lactuca sativa) by integrating Internet of Things (IoT), automated nutrient management, and image-based plant monitoring. An ESP32 is used as the main controller to acquire data from sensors measuring air temperature and humidity, nutrient-solution temperature, pH, electrical conductivity/total dissolved solids (EC/TDS), and reservoir water level. The system automates nutrient-solution circulation using a 12 V DC water pump and controls nutrient dosing through two mini peristaltic dosing pumps dedicated to Nutrient A and Nutrient B. A full-spectrum LED grow light provides artificial illumination to support indoor lettuce growth. In addition, a USB camera is positioned facing the plants to capture lettuce leaf images. A Raspberry Pi 5 performs image processing using OpenCV to analyse leaf colour and leaf size as indicators of plant growth and condition. Sensor readings and processed information are communicated using MQTT and integrated with Node-RED for real-time monitoring and dashboard visualisation. The developed system aims to reduce dependence on manual monitoring by integrating environmental sensing, automated water and nutrient management, and image-based plant observation within a compact vertical hydroponic system.

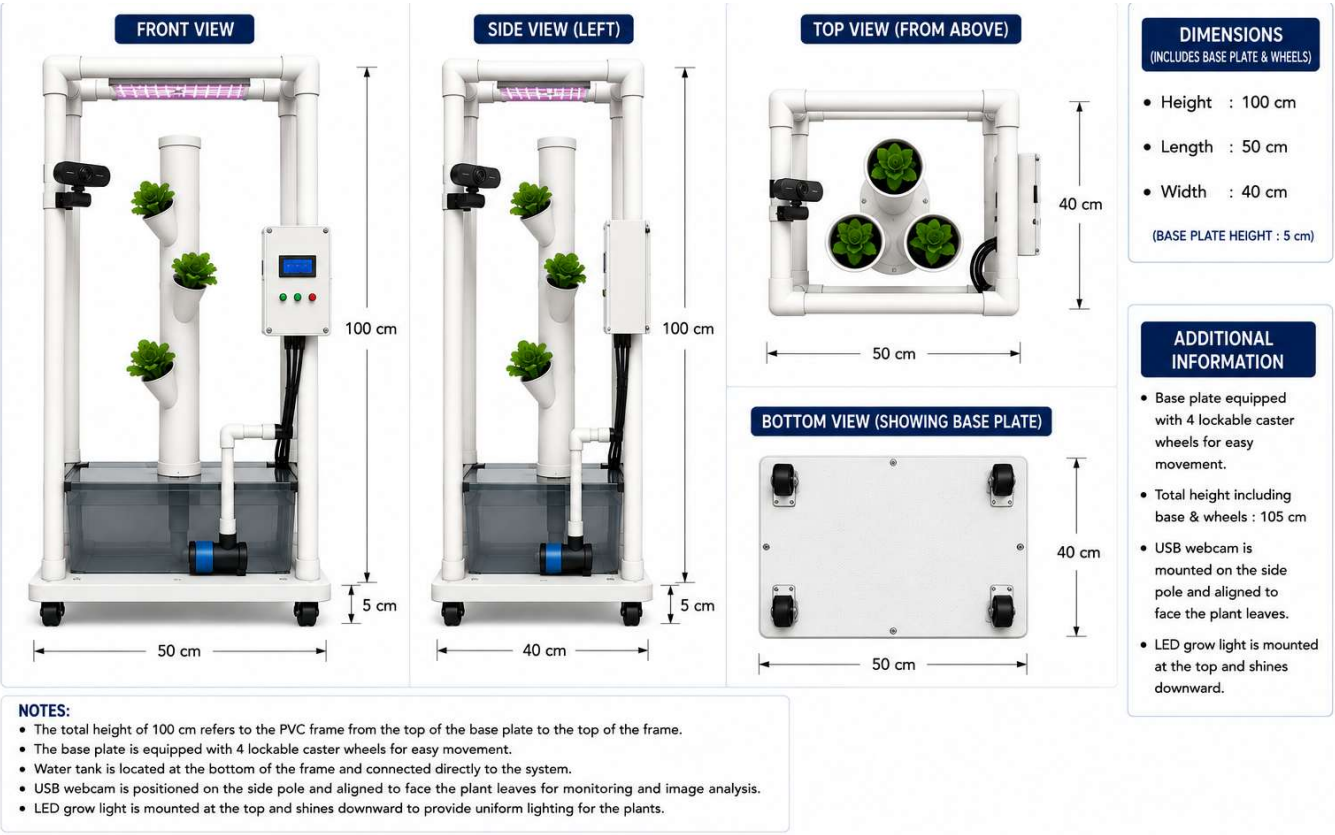
Keywords: Indoor vertical garden, hydroponics, Lactuca sativa, IoT, ESP32, Raspberry Pi 5, OpenCV, Node-RED, nutrient dosing, automation.

ABSTRAK

Penanaman secara menegak di dalam bangunan merupakan satu pendekatan praktikal bagi penanaman tanaman dalam ruang yang terhad. Walau bagaimanapun, keadaan persekitaran dan larutan nutrien perlu dipantau secara berterusan serta dikawal dengan sewajarnya. Projek ini mencadangkan Sistem Pemantauan dan Automasi Taman Menegak Dalaman untuk penanaman salad (Lactuca sativa) secara hidroponik dengan mengintegrasikan Internet Benda (IoT), pengurusan nutrien secara automatik dan pemantauan tanaman berasaskan imej. ESP32 digunakan sebagai pengawal utama untuk memperoleh data daripada sensor yang mengukur suhu dan kelembapan udara, suhu larutan nutrien, pH, kekonduksian elektrik/jumlah pepejal terlarut (EC/TDS), serta paras air dalam takungan. Sistem mengautomasikan peredaran larutan nutrien menggunakan pam air DC 12 V dan mengawal pemberian nutrien melalui dua pam dos peristaltik mini yang dikhaskan untuk Nutrien A dan Nutrien B. Lampu LED grow light spektrum penuh digunakan untuk menyediakan pencahayaan buatan bagi menyokong pertumbuhan salad di dalam bangunan. Selain itu, kamera USB dihalakan ke arah tanaman untuk mengambil imej daun salad. Raspberry Pi 5 menjalankan pemprosesan imej menggunakan OpenCV bagi menganalisis warna dan saiz daun sebagai petunjuk kepada pertumbuhan dan keadaan tanaman. Bacaan sensor dan maklumat yang telah diproses dihantar menggunakan MQTT dan diintegrasikan dengan Node-RED untuk pemantauan masa nyata serta visualisasi melalui papan pemuka. Sistem yang dibangunkan bertujuan mengurangkan kebergantungan terhadap pemantauan manual dengan menggabungkan penderiaan persekitaran, pengurusan air dan nutrien secara automatik serta pemerhatian tanaman berasaskan imej dalam sebuah sistem hidroponik menegak yang kompak.

Kata kunci: Taman menegak dalaman, hidroponik, Lactuca sativa, IoT, ESP32, Raspberry Pi 5, OpenCV, Node-RED, dos nutrien, automasi.

PROTOTYPE DESIGN



Prototype Dimensions

Part	Dimension / Placement
PVC frame	50 cm (W) × 40 cm (D) × 100 cm (H)
PVC planting tower	80 cm height × 4-inch diameter (10.16 cm)
Planting level 1	20 cm from top
Planting level 2	40 cm from top
Planting level 3	60 cm from top
Vertical plant spacing	20 cm
Net-pot arrangement	3 ports: Left → Right → Left, angled
Base plate	50 cm x 40 cm; mounted directly below the water tank within the prototype footprint
Lockable caster wheels	4 units; mounted beneath the four corners of the base plate
Transparent nutrient reservoir	29 cm × 20 cm × 15.7 cm
USB camera	Mounted close to and facing lettuce leaves
Ultrasonic sensor	Mounted above reservoir, facing downward
Dosing pumps	2 units behind reservoir, outside water tank

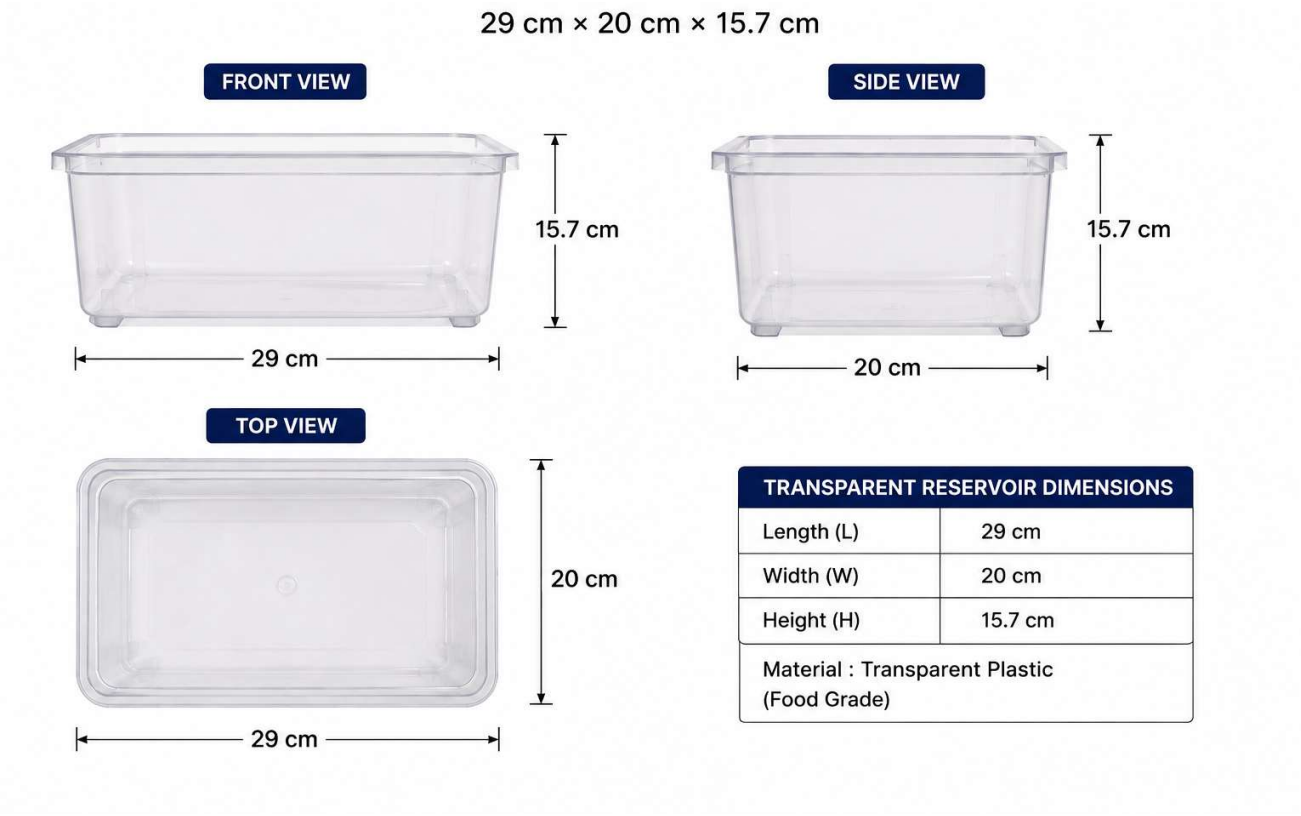
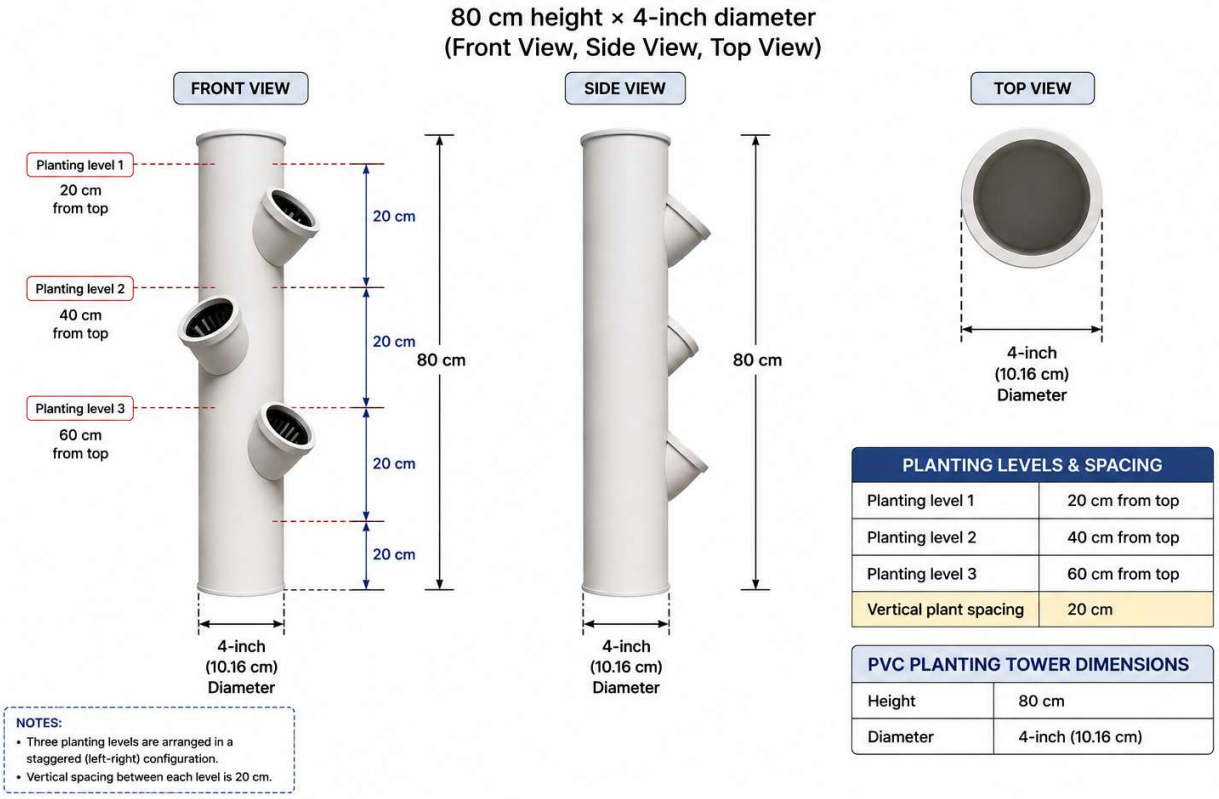


Figure: Transparent reservoir dimensions - Front, Side and Top View

PVC PLANTING TOWER DIMENSIONS



PLANTING LEVELS & SPACING	
Planting level 1	20 cm from top
Planting level 2	40 cm from top
Planting level 3	60 cm from top
Vertical plant spacing	20 cm

PVC PLANTING TOWER DIMENSIONS	
Height	80 cm
Diameter	4-inch (10.16 cm)

Figure: PVC planting tower - 80 cm height, 4-inch diameter, planting levels at 20 cm, 40 cm and 60 cm

PVC FRAME DIMENSIONS

50 cm (W) × 40 cm (D) × 100 cm (H)



FRAME SPECIFICATIONS

Width (W)	50 cm
Depth (D)	40 cm
Height (H)	100 cm
Base Plate Size	50 cm (W) × 40 cm (D)
Base Plate Thickness	1.0 cm (approx.)
Wheels	4 Lockable Caster Wheels
Material	PVC Pipe (White) & PVC Fittings

NOTES

- The total height of 100 cm refers to the PVC frame from the top of the base plate to the top of the frame.
- The base plate is located at the bottom of the frame (below the water tank) and equipped with 4 lockable caster wheels for easy movement.
- Base plate and wheels are positioned inside the footprint of the PVC frame (50 cm × 40 cm).
- All components (water tank, pumps, sensors, control box, camera, etc.) are mounted on the frame as shown in the overall design.

Figure: PVC frame dimensions - 50 cm (W) x 40 cm (D) x 100 cm (H)

SCHEMATIC DIAGRAM

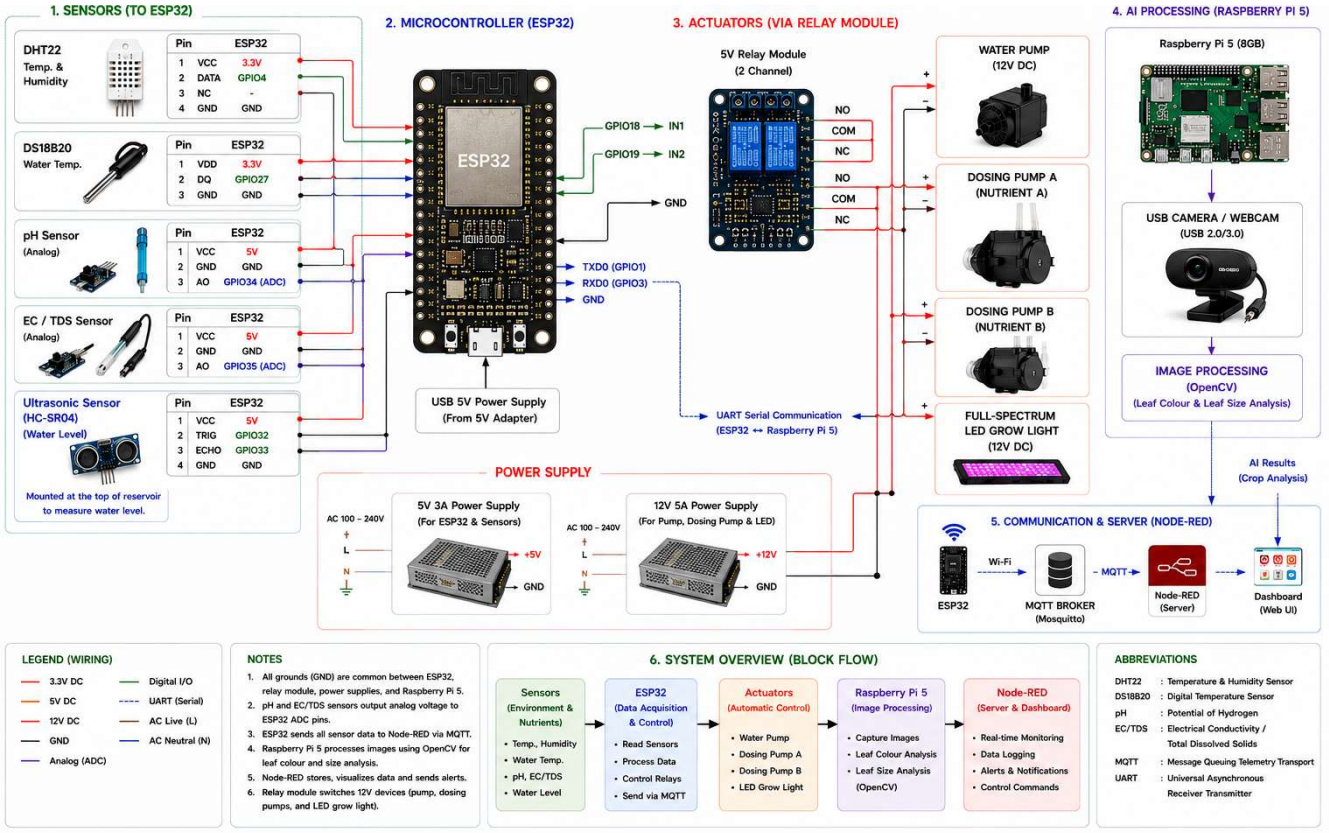


Figure: Indoor Vertical Garden Monitoring and Automation System - Schematic Diagram

COMPONENT LIST

Sensors

No.	Component	Qty	Function
1	DHT22 Temperature & Humidity Sensor	1	Ambient air temperature and humidity
2	DS18B20 Waterproof Temperature Sensor	1	Nutrient-solution temperature
3	pH Sensor + Probe	1	Nutrient-solution pH
4	EC/TDS Sensor + Probe	1	Nutrient concentration / conductivity
5	Ultrasonic Sensor (HC-SR04)	1	Reservoir water level; faces downward

Controller, Processing & Camera

No.	Component	Qty	Function
6	ESP32 Development Board	1	Reads sensors and controls actuators
7	Raspberry Pi 5	1	Local processing and image analysis
8	USB Camera / Webcam	1	Captures lettuce leaf colour and size

Actuators & Automation

No.	Component	Qty	Function
9	12 V DC Water Pump	1	Circulates nutrient solution to tower
10	GROTHEN DC 12 V Mini Peristaltic Dosing Pump	2	Nutrient A and Nutrient B dosing
11	Full-Spectrum LED Grow Light	1	Artificial light for lettuce
12	Relay Module	1	ON/OFF actuator switching

Nutrient Dosing System

No.	Component	Qty	Function
13	Nutrient A Bottle	1	Stores Nutrient A
14	Nutrient B Bottle	1	Stores Nutrient B
15	Silicone Dosing Tubing	As required	Dosing lines to reservoir

Power & Electrical

No.	Component	Qty	Function
16	12 V DC Power Supply	1	Power for 12 V loads
17	5 V / 5 A Raspberry Pi 5 Power Supply	1	Dedicated Raspberry Pi 5 supply
18	DC-DC Buck Converter / Regulator	1	Regulated voltage where required
19	Terminal Block / Power Distribution	1 set	Electrical distribution
20	Fuse + Fuse Holder	1 set	Electrical protection
21	Main Power Switch	1	Main system ON/OFF
22	USB / Power Cables + Electrical Wiring	As required	Power, signal and communication connections
23	Cable Glands / Connectors	As required	Cable protection and enclosure entry

Structure & Hydroponic Components

No.	Component	Qty	Function
24	PVC Planting Tower – 4 inch	1	80 cm vertical planting tower
25	Net Pot	3	Holds lettuce and growing medium
26	PVC Frame + Fittings	1 set	50 × 40 × 100 cm support frame
27	Transparent Nutrient Reservoir / Water Tank	1	29 × 20 × 15.7 cm reservoir
28	Flexible Water Tubing / Hose	As required	Water circulation line
29	Electronic Control Box	1	Houses ESP32, Raspberry Pi 5, relay, converter and wiring
30	Base Plate	1	Supports the complete prototype and provides mounting surface for caster wheels
31	Lockable Caster Wheel	4	Allows easy movement and locks the prototype securely in position

Software & Communication

- Node-RED — monitoring and dashboard visualisation
- MQTT Broker — system data communication
- Python — system and image-processing programming
- OpenCV — lettuce leaf colour and size analysis